

CH5440W Assignment 2 Part 1

Individual Submissions; Submit a neat copy (handwritten or typed)

- a) **Write the question number and subdivisions before the answer**
- b) *Suggested to do numerical problems with Matlab/Excel/Python/JMP – Your choice*
- c) Mention all references used prominently in the answers
- d) **Deadline: 10th August 2026**

Question 1

[10+5+5]

We have learnt about PLS2 using NIPALS algorithm in the classes.

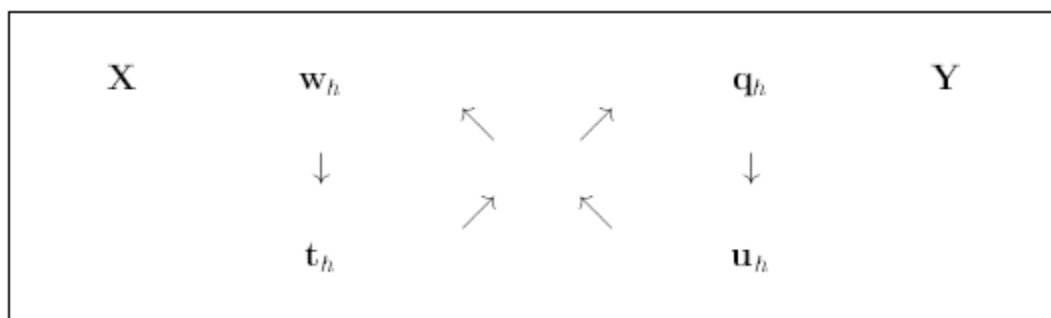
- a) Discuss next on PLS1 algorithm using SIMPLS algorithm.
- b) Bring out the difference between the two approaches
- c) Explain where each approach is preferred/applied.

Question 2

[4+2+2+2+2+2+3+3+3]

With reference to the NIPALS algorithm dealt in class, answer the following questions in your own words

- a) What is the difference between weights and loading?
- b) Indicate how X and Y scores condense voluminous data involving multiple features
- c) What is meant by deflation/reconstruction?
- d) Explain the figure given below (reference to the diplomat thesis given in class)



- e) To find weights $\mathbf{w}$ why do you use Y-scores $\mathbf{u}$ rather than $\mathbf{t}$?
- f) Why do you normalize the weights $\mathbf{w}$ using its norm?
- g) To find X scores, Why regress E'_{h-1} with $w_h t'_h$? Should it not be $p_h t'_h$?
- h) Explain the following steps in NIPALS PLS2 algorithm

Step 9: Fit $\mathbf{t}_h$ to the newly gained $\mathbf{p}_h$ as $t_h = t_h \|\mathbf{p}_h\|$

Step 10: Normalize $\mathbf{p}'_h$ as $p'_h = \frac{p'_h}{\|\mathbf{p}_h\|}$

- i) Why find by regression weights $\mathbf{w}$ for X residuals but loadings $\mathbf{q}$ for Y residuals?

$$E_{h-1} = u_h w'_h + \quad F_{h-1} = t_h q'_h +$$

Question 3

[4+6+4]

- a) For facilitating your coding with Python and for a general understanding of the code's workflow, neatly summarize the PLS2 algorithm step by step.
- b) Solve the following problem without centering or scaling or standardizing
- c) Report PRESS and root mean square PRESS
- d) Compare and validate all your Matlab/Python answers with JMP (**W,T,U,Q,P,Beta**, diagonal matrix **B**, Root Mean Square PRESS). In JMP use "leave one out" option.
- e) Show finally how the PLS2 led to predictions of the outcomes $\mathbf{Y}$.

Note: Use 3 factors

```

X = [
    2    5    3    6    8    1
    4    6    5    7    9    2
    5    8    6    8   10    3
    7    9    8   10   12    5
    9   11    9   12   13    6
];

```

```

Y = [
    20  35
    24  40
    28  45
    36  58
    42  66
];

```

Question 4

[10]

For the big data set (Gasoline.xls) carry out PLS2 analysis and find the model coefficients beta, **W**, **T**, **U**, **Q**, **P** and **B**.